Part A

Question 1:

a)

$$1573 = 11^2 \times 13$$

Using Proposition 2.3, HB chapter 2 p17;

for $n \geq 2$, $n = p_1^{k_1} p_2^{k_2} \dots p_r^{k_r}$, then

$$\tau(n) = (k_1 + 1)(k_2 + 1) \dots (k_r + 1) = 1573 = 11^2 \times 13$$

Where $\tau(n)$ is the number of positive divisors of n.

$$k_1 + 1 = 13 \implies k_1 = 12$$

$$k_2 + 1 = 11 \implies k_2 = 10$$

$$k_3 + 1 = 11 \implies k_3 = 10$$

Any other allocation either increases the largest exponent or introduces larger primes, producing a larger integer.

To minimise n, we assign the largest exponent to the smallest prime, the second largest exponent to the second smallest prime, and so on.

Since $1573 = 13 \cdot 11 \cdot 11$, we choose exponents $12, 10, 10$

Hence the smallest positive integer n with exactly 1573 positive divisors is

$$\boxed{n = 2^{12} \cdot 3^{10} \cdot 5^{10}}$$

b)
Let n be the positive integer such that

$$7n + \sigma(n) \mid 8$$

Where $\sigma(n)$ is the sum of divisors function.

If $n = p^2$ (where p is prime), then $p \equiv 7 \pmod 8$

Proof.

The divisors of p^2 are $1, p$ and p^2

Using Proposition 1.2, HB Chapter 9 p48;

$$\sigma(n) = \sum_{d|n} d$$

So

$$\sigma(p^2) = 1 + p + p^2$$

As

$$7n + \sigma(n) \equiv 0 \pmod 8$$

Substituting $n = p^2$ and $\sigma(p^2)$ gives

$$7p^2 + \sigma(p^2) \equiv 0 \pmod 8$$

$$7p^2 + 1 + p + p^2 \equiv 0 \pmod 8$$

Simplifying

$$8p^2 + p + 1 \equiv 0 \pmod 8$$

As $8p^2 \equiv 0 \pmod 8$, since $8 \mid 8p^2$, we have

$$p + 1 \equiv 0 \pmod 8$$

$$p \equiv -1 \pmod 8 \Rightarrow p \equiv 7 \pmod 8$$

$\square$

Question 2:

$$2x^2 + 3x + 4 \equiv 0 \pmod 7$$

a)
i.
Using Euler's criterion, HB Chapter 10 p45, we calculate the discriminant,

$$\begin{aligned} b^2 - 4ac &= 3^2 - 4 \cdot 2 \cdot 4 \\ &= 9 - 32 \\ &= -23 \\ &\equiv 5 \pmod 7 \end{aligned}$$

Since the congruence has solutions if and only if the discriminant is a quadratic residue modulo 7, we apply Euler's Criterion.

Using Theorem 1.5, HB Chapter 10 pg 45, we check if 5 is a quadratic residue $\pmod 7$.

$$\begin{aligned} a^{(p-1)/2} &\equiv 1 \pmod p \\ 5^{(7-1)/2} &\equiv 5^3 \pmod 7 \\ &\equiv 125 \pmod 7 \\ &\equiv 6 \pmod 7 \\ &\equiv -1 \pmod 7 \end{aligned}$$

So 5 is a quadratic non-residue $\pmod 7$, and therefore the congruence

$$2x^2 + 3x + 4 \equiv 0 \pmod 7$$

has no solutions.

ii.

$$2x^2 + 3x + 4 \equiv 0 \pmod 7$$

Find the discriminant:

$$\begin{aligned} b^2 - 4ac &= 3^2 - 4 \cdot 2 \cdot 4 \\ &\equiv 9 - 32 \pmod 7 \\ &\equiv 9 + 3 \pmod 7 \\ &\equiv 12 \pmod 7 \\ &\equiv 5 \pmod 7 \end{aligned}$$

By Gauss's lemma, Theorem3.1 Chapter 10 p46 we determine whether 5 is a quadratic residue modulo 7.:

We need the first $\frac{p-1}{2} = 3$ multiples of $5 \pmod 7$:

$$\begin{aligned} 1 \cdot 5 &\equiv 5 \pmod 7 \\ 2 \cdot 5 &\equiv 3 \pmod 7 \\ 3 \cdot 5 &\equiv 1 \pmod 7 \end{aligned}$$

$$S = \{5, 3, 1\} \pmod 7$$

Looking for the values greater than $\frac{p}{2}$, hence $n = 1$

Using Legendre symbol, Definition 2.1 Chapter10 p45, and Gauss's lemma:

$$\left(\frac{5}{7}\right) = (-1)^1 = -1$$

Thus there are no solutions to the quadratic congruence $2x^2 + 3x + 4 \equiv 0 \pmod 7$.

b)

i.

$$\left(\frac{-48}{229}\right)$$

Theorem 2.2(c), HB Chapter 10 p46

$$\left(\frac{-48}{229}\right) = \left(\frac{-1}{229}\right) \cdot \left(\frac{16}{229}\right) \cdot \left(\frac{3}{229}\right)$$

Calculating each term separately, using standard results

$$\left(\frac{-1}{229}\right) = 1$$

Theorem 2.2(e), HB Chapter 10 p46

$$\left(\frac{16}{229}\right) = 1$$

Theorem 2.2(b), HB Chapter 10 p46

$$\left(\frac{3}{229}\right) = 1$$

Proposition 4.7, HB Chapter 10 p46

Combining these results

$$\left(\frac{-48}{229}\right) = 1 \cdot 1 \cdot 1$$

$$= 1$$

So -48 is a quadratic residue modulo 229

ii.

$$\left(\frac{151}{43}\right)$$

$$\left(\frac{151}{43}\right) = \left(\frac{22}{43}\right)$$

$$= \left(\frac{2}{43}\right) \cdot \left(\frac{11}{43}\right)$$

Theorem 2.2(c), HB Chapter 10 p46

Calculating each term separately

$$\left(\frac{2}{43}\right) = -1$$

Proposition 3.2, HB Chapter 10 p46

$$\left(\frac{11}{43}\right) = -\left(\frac{43}{11}\right)$$

Theorem 4.2, HB Chapter 10 p46

$$= -\left(\frac{10}{11}\right)$$

$$= -\left(\frac{2}{11}\right) \cdot \left(\frac{5}{11}\right)$$

Theorem 2.2(c), HB Chapter 10 p46

Calculate each term seperatly

$$\left(\frac{2}{11}\right) = -1$$

Proposition 3.2, HB Chapter 10 p46

$$\left(\frac{5}{11}\right) = \left(\frac{11}{5}\right)$$

Theorem 4.2, HB Chapter 10 p46

$$= \left(\frac{1}{5}\right)$$

$$\left(\frac{1}{5}\right) = 1$$

Theorem 2.2(b), HB Chapter 10 p46

Combining these results

$$\left(\frac{151}{43}\right) = \left(\frac{22}{43}\right) = \left(\frac{2}{43}\right)\left(\frac{11}{43}\right)$$

$$\left(\frac{2}{43}\right) = -1$$

and

$$\left(\frac{11}{43}\right) = -\left(\frac{10}{11}\right) = -\left(\frac{2}{11}\right) \cdot \left(\frac{5}{11}\right)$$

$$\left(\frac{2}{11}\right) = -1$$

$$\left(\frac{5}{11}\right) = 1$$

So

$$\left(\frac{11}{43}\right) = -\left(\frac{2}{11}\right) \cdot \left(\frac{5}{11}\right) = -((-1)(1)) = 1$$

therefore

$$\left(\frac{151}{43}\right) = \left(\frac{2}{43}\right)\left(\frac{11}{43}\right) = (-1)(1) = -1$$

So 151 is a quadratic non-residue modulo 43

Question 3:

$$f(x) = 3x^5 + 11x^4 + 4x^3 - 8x^2 - 6x$$

and

$$g(x) = 3x^3 + 11x^2 + 7x + 3$$

Using Euclidean algorithm to find $\text{hcf}(f(x), g(x))$

Theorem 3.1, HB Chapter 11 p50

$$f(x) \mid g(x) = q(x) \cdot g(x) + r(x)$$

$$
\begin{array}{rr}
 & x^2 \qquad\quad -1 \\ \cline{2-2}
3x^3+11x^2+7x+3\big) & 3x^5+11x^4+4x^3 \quad -8x^2-6x \quad \\
 & -3x^5-11x^4-7x^3 \quad -3x^2 \qquad\qquad \\ \cline{2-2}
 & -3x^3-11x^2-6x \quad \\
 & 3x^3+11x^2+7x+3 \\ \cline{2-2}
 & x+3
\end{array}
$$

$$f(x) = (x^2 - 1) \cdot (3x^3 + 11x^2 + 7x + 3) + (x + 3)$$

$$g(x) \mid r(x) = q(x) \cdot r(x) + s(x)$$

$$
\begin{array}{rr}
 & 3x^2+2x+1 \\ \cline{2-2}
x+3\big) & 3x^3+11x^2+7x+3 \\
 & -3x^3 \quad -9x^2 \qquad\qquad\quad \\ \cline{2-2}
 & 2x^2+7x \quad\;\; \\
 & -2x^2-6x \quad\;\; \\ \cline{2-2}
 & x+3 \\
 & -x-3 \\ \cline{2-2}
 & 0
\end{array}
$$

$$g(x) = (3x^2 + 2x + 1)(x + 3) + 0$$

The remainder is 0, so we stop here. The last non-zero remainder is $x + 3$ which is already monatic.

Therefore the highest common factor is,

$$\text{hcf}(f, g) = x + 3$$

Question 4:

$$f(x) = x^6 + 4x^3 - 2x^2 + 6x + 2$$

a)
Using the Rational Root test, HB Chapter 11 p52, we check for possible rational roots.

$$p \mid 2 \quad q \mid 1$$

The possible rational roots are $\frac{p}{q} = \pm 1, \pm 2$.

$$f(1) = 1 + 4 - 2 + 6 + 2 = 11 \neq 0$$
$$f(-1) = 1 - 4 - 2 - 6 + 2 = -9 \neq 0$$
$$f(2) = 64 + 32 - 8 + 12 + 2 = 102 \neq 0$$
$$f(-2) = 64 - 32 - 8 - 12 + 2 = 10 \neq 0$$

Therefore $f(x)$ has no rational roots. Hence $f(x)$ has no linear factor over $\mathbb{Q}$.

b)
Using Eisenstein's criterion, HB Chapter 11 p52, we check for irreducibility over $\mathbb{Q}$.

Let $p = 2$.

- $2 \mid 4, -2, 6, 2$
- $2 \nmid 1$
- $2^2 = 4 \nmid 2$

So by Eisenstein's criterion, $f(x)$ is irreducible over $\mathbb{Q}$, hence it has no linear factor over $\mathbb{Q}$, so it has no rational roots.

Question 5:

a)
i.
Suppose;

$$\sqrt{50} = \frac{m}{n}, \quad wherem, n \in \mathbb{Z}^+$$

Show

$$\sqrt{50} = \frac{50n - 7m}{m - 7n}$$

Proof.

Since

$$\sqrt{50} = \frac{m}{n}$$

We have

$$m = n\sqrt{50}$$

$$\frac{m}{n} = \frac{50n - 7m}{m - 7n}$$

Cross-multiplying

$$m(m - 7n) = n(50n - 7m)$$

expanding

$$m^2 - 7mn = 50n^2 - 7mn$$

Adding $7mn$ to both sides

$$m^2 = 50n^2$$

Taking square roots of both sides

$$\sqrt{m^2} = \sqrt{50n^2}$$

$$m = n\sqrt{50}$$

Dividing both sides by n

$$\frac{m}{n} = \sqrt{50}$$

$$\therefore \sqrt{50} = \frac{m}{n}$$

□

ii.
Suppose;

$$\sqrt{50} = \frac{m}{n} = \frac{50n - 7m}{m - 7n}$$

By Fermat's infinite descent, HB Chapter12 p53, to show that $\sqrt{50}$ is irrational.

Proof.

Assuming $\sqrt{50}$ is rational, then $\frac{m}{n}$ is in lowest terms
With $m, n \in \mathbb{Z}^+$ and $\operatorname{hcf}(m, n) = 1$
From part (a), we have

$$\sqrt{50} = \frac{50n - 7m}{m - 7n}$$

We define $m' = 50n - 7m, \quad n' = m - 7n$
Since $m, n \in \mathbb{Z}$, it follows that $m', n' \in \mathbb{Z}$
Since

$$7 < \sqrt{50} < 8$$

$$m' = 50n - 7m = n\left(50 - 7\sqrt{50}\right) > 0$$

and

$$n' = m - 7n = n\left(\sqrt{50} - 7\right) > 0$$

So $m', n' \in \mathbb{Z}^+$
Also, using $7 < \sqrt{50}$

$$m' = n\left(50 - 7\sqrt{50}\right) < n\,(50 - 49) = n < m$$

and since $m = n\sqrt{50} > n, \quad n' < n$

$$\frac{m'}{n'} = \frac{m}{n}$$

but with $m' < m$ and $n' < n$, this contradicts our assumption that $\frac{m}{n}$ is in lowest terms.

Repeating this process would produce a infinite decreasing sequence of positive integers which is impossible. So Our initial assumption leads to a contradiction.

Therefore $\sqrt{50}$ is irrational.

$\square$

b)
i.

$$578 = 2 \times 289$$

since;

$$289 = 17^2$$

So we get our first representation as the sum of two squares;

$$578 = 17^2 + 17^2$$

$$23^2 = 529$$

Subtracting from 578 gives;

$$578 - 23^2 = 49$$

$$49 = 7^2$$

So our second representation as the sum of two squares is;

$$578 = 23^2 + 7^2$$

Thus the two representations of 578 as the sum of two squares are: First representation:

$$\boxed{578 = 17^2 + 17^2}$$

Second representation:

$$\boxed{578 = 23^2 + 7^2}$$

ii.

$$1715 = 2 \times 7^3$$

$$7 \equiv 3 \pmod 4$$

Using Proposition 4.13 Chapter 12 p57, since $7 \equiv 3 \pmod 4$ appears with odd exponent, 1715 cannot be expressed as the sum of two squares.

Question 6:

$R = \{a_0 + a_1x + a_2x^2 + a_3x^3 + \ldots + a_nx^n : a_0, a_1, \ldots, a_n \in \mathbb{Z}_{59},\ a_1 = 0,\ n \geq 0\}$

a)

Proof. Using the subring criterion, Lemma 1.8 HB Chapter 11 p48,

Closure under subtraction;

Let

$$f(x) = a_0 + a_2x^2 + a_3x^3 \ldots$$

$$g(x) = b_0 + b_2x^2 + b_3x^3 \ldots$$

Then

$$f(x) - g(x) = (a_0 - b_0) + (a_2 - b_2)x^2 + (a_3 - b_3)x^3 \ldots$$

Since $a_i, b_i \in \mathbb{Z}_{59}$, then $a_i - b_i \in \mathbb{Z}_{59}$

Also, the coefficient of x is $0 - 0 = 0$

So $f(x) - g(x) \in R$

Closure under multiplication;

In order to show closure under multiplication of a polynomial with no x term we consider the product of multiplying two polynomials in R to create a x term.

The coefficient of x in a product comes from terms where the degrees add to 1. This can happen with terms of degree $1 + 0$ or $0 + 1$.

As the coefficient of x in both polynomials is 0, the coefficient of x in the product will also be 0.

Thus the coefficient of x in $f \cdot g$ is 0 so $f \cdot g \in R$.

Since $R \subseteq \mathbb{Z}_{59}[x]$, we apply the subring criterion.

Hence, R is a subring of $\mathbb{Z}_{59}[x]$.

□

Since $1 = 1 + 0x + 0x^2 + \ldots \in R$, R contains the multiplicative identity.

Units in a polynomial ring over a field are the non-zero constants, Definition 1.14 Chapter 11 p48.

If $u \in R$ is a unit in R, then it is alson a unit in $\mathbb{Z}_{59}[x]$ hence must be a non-zero constant, so $U(R) = \mathbb{Z}_{59}^*$

Therefore the units in R are given by;

$$U(R) = \{a_0 : a_0 \in \mathbb{Z}_{59}^*\}$$

where

$$\mathbb{Z}_{59}^* = \{1, 2, 3, \ldots, 58\}$$

b)

By Lemma 2.6 chapter 11 p49,

In order for x^2 to be reducible in R there must exist polynomials $p(x), q(x) \in R$ such that

$$x^2 = p(x) \cdot q(x)$$

This leave three possibilities for the degrees of $p(x)$ and $q(x)$:

- $\deg(p) = 0, \deg(q) = 2$
- $\deg(p) = 2, \deg(q) = 0$
- $\deg(p) = 1, \deg(q) = 1$

The first two possibilities are not valid as the constant polynomials are units in R and hence cannot give a non-trivial factorisation.

The third possibility is not valid since no polynomial in R has a non-zero coefficient of x.

c)

$$x^2 \nmid x^3 \text{ in } R$$

Proof. Assume for contradiction that $x^2 \mid x^3$ in R. Then there exists $h(x) \in R$ such that

$$x^3 = x^2 \cdot h(x).$$

In $\mathbb{Z}_{59}[x]$, this forces $h(x) = x$. But $x \notin R$ because elements of R have zero coefficient of x. This contradiction shows $x^2 \nmid x^3$ in R. □

Prove x^2 is not a prime element in R.

Proof.

For x^2 to be prime in R, whenever $x^2 \mid f(x) \cdot g(x)$, then $x^2 \mid f(x)$ or $x^2 \mid g(x)$.

Consider the polynomials

$$x^2 \mid x^6 \text{ since } x^6 = x^2 \cdot x^4 \text{ both } x^4, x^6 \in R$$

But

$$x^6 = x^3 \cdot x^3 \text{ and } x^2 \nmid x^3$$

An element p is prime if whenever $p \mid ab$ then $p \mid a$ or $p \mid b$

Hence x^2 is not a prime element in R. □

Question 7:

PA Paul Allen
25 Dec 2025 23:34

I have been going through this question and think I can solve it using algebra. But I have been looking ahead in the work and it seems there is possibly other ways to solve part a) but it isn't covered until book E. I haven't gone through that work properly yet so I'm not sure if I am even correct, I was wondering if we are expected to be using stuff ahead of where we are at?

Has anybody else found alternative solutions to previously done questions as we get further into the material?

Any advise would be welcome before I get ahead of myself.

Thanks

Paul

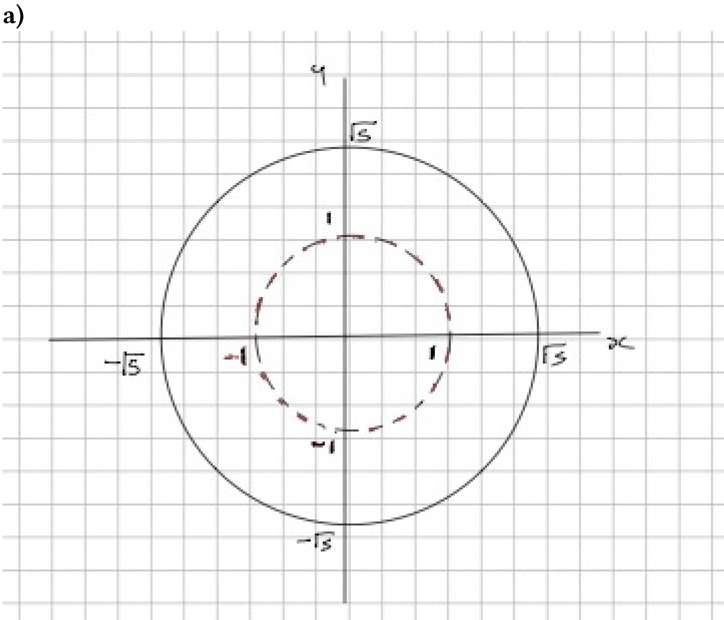
Part B

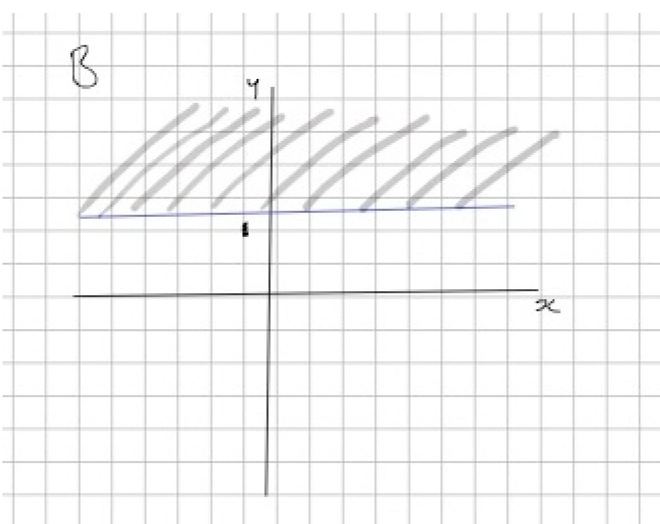
Question 8:

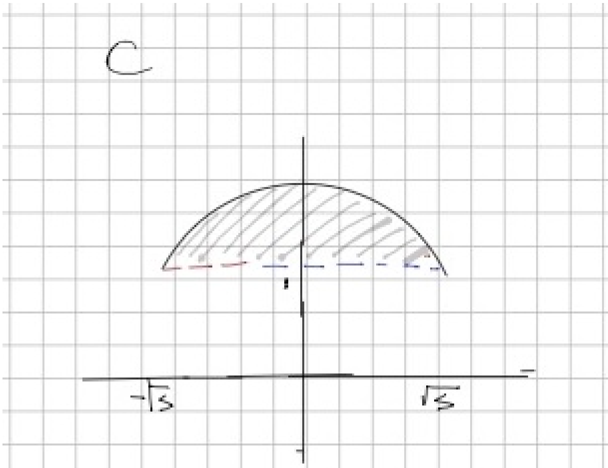
Question 8 from the TMA03 assignment.

$$A = \{(x, y) \in \mathbb{R}^2 : 1 < x^2 + y^2 \leq 5\}$$
$$B = \{(x, y) \in \mathbb{R}^2 : y \geq 1\}$$
$$C = A \cap B$$

a)







b)

$$\left((0, 1+\frac{1}{n})\right)_{n=1}^{\infty}$$

By Definition 3.4 HB Chapter 13 p67,

The first component is a constant sequence $\lim_{n\to\infty} = 0$ and constants sequences converge to the constant value.

The second component is $1 + \frac{1}{n}$.

Splitting this into two limits;

$$\lim_{n\to\infty} 1 = 1$$

Again it is a constant sequence, so it converges to 1.

$$\lim_{n\to\infty} \frac{1}{n} = 0$$

As n *approaches infinity,* $\frac{1}{n}$ *approaches* 0.

Combining these results;

$$\lim_{n\to\infty}(1+\frac{1}{n}) = 1+0 = 1$$

Therefore, by Definition 3.4 HB Chapter 13 p67, the sequence

$$\left((0, 1+\frac{1}{n})\right)_{n=1}^{\infty}$$

converges to

$$\boxed{(0,1)}$$

To test the limit point $(0,1) \in A$,

$$1 < x^2 + y^2 \leq 5$$

$$1 < 0^2 + 1^2 \leq 5$$

$$1 < 1 \leq 5$$

This is false, so $(0,1) \notin A$

To test the limit point $(0,1) \in B$,

$$y \geq 1$$

$$1 \geq 1$$

This is true, so $(0,1) \in B$

Hence limit point $(0,1) \notin A$ but $(0,1) \in B$,

Therefore $(0,1) \notin C$.

Question 9:

Question 10 from the TMA03 assignment.

Let

$$R = (\mathbb{Z}, \oplus, \odot)$$

Where

$$\oplus = a + b + 1$$

and

$$\odot = ab + a + b$$

a)

Proof.

R1 and R6 Closure of $\oplus$ and $\odot$;

$$a, b, 1 \in \mathbb{Z}$$

Therefore„ because of closer under addition in $\mathbb{Z}$,

$$a + b + 1 \in \mathbb{Z}$$

So $a \oplus b \in \mathbb{Z}$

$$ab, a, b \in \mathbb{Z}$$

Therefore, because of closer under multiplication and addition in $\mathbb{Z}$,

$$ab + a + b \in \mathbb{Z}$$

So $a \odot b \in \mathbb{Z}$

R2 and R7 Associativity of $\oplus$ and $\odot$;

For $\oplus$,

$$(a \oplus b) \oplus c = (a + b + 1) + c + 1$$

$$= a + b + c + 2$$

And

$$a \oplus (b \oplus c) = a + (b + c + 1) + 1$$

$$= a + b + c + 2$$

So $(a \oplus b) \oplus c = a \oplus (b \oplus c)$

For $\odot$,

$$(a \odot b) \odot c = (ab + a + b) \odot c$$

$$= (ab + a + b)c + (ab + a + b) + c$$

$$= abc + ac + bc + ab + a + b + c$$

And

$$a \odot (b \odot c) = a \odot (bc + b + c)$$

$$= a(bc + b + c) + a + (bc + b + c)$$

$$= abc + ab + ac + a + bc + b + c$$

So $(a \odot b) \odot c = a \odot (b \odot c)$

R3 Additive identity;

Let e be the additive identity, then

$$a \oplus e = a$$

$$a + e + 1 = a$$

$$e + 1 = 0$$

$$e = -1$$

So the additive identity is -1

R8 Multiplicative identity;

Let f be the multiplicative identity, then

$$a \odot f = a$$

$$af + a + f = a$$

$$af + f = 0$$

$$f(a+1) = 0$$

For a given a this can happen if $f = 0$ or $a = -1$

But f must work for all a, and we can choose $a \neq -1$ e.g. $a = 0$

$$a + 1 \neq 0$$

$$0 + 1 \neq 0$$

$$1 \neq 0$$

Hence $f = 0$

R4 Additive inverse;

Let g be the additive inverse of a, then

$$a \oplus g = -1$$

$$a + g + 1 = -1$$

$$g + a = -2$$

$$g = -a - 2$$

So the additive inverse of a is $-a - 2$

R5 Commutativity of $\oplus$;

$$\begin{aligned} a \oplus b &= a + b + 1 \\ &= b + a + 1 \\ &= b \oplus a \end{aligned}$$

So $a \oplus b = b \oplus a$

R9 Distributive law;

$$\begin{aligned} a \odot (b \oplus c) &= a \odot (b + c + 1) \\ &= a(b + c + 1) + a + (b + c + 1) \\ &= ab + ac + a + a + b + c + 1 \\ &= ab + ac + 2a + b + c + 1 \end{aligned}$$

And

$$\begin{aligned} (a \odot b) \oplus (a \odot c) &= (ab + a + b) \oplus (ac + a + c) \\ &= (ab + a + b) + (ac + a + c) + 1 \\ &= ab + ac + a + a + b + c + 1 \\ &= ab + ac + 2a + b + c + 1 \end{aligned}$$

So $a \odot (b \oplus c) = (a \odot b) \oplus (a \odot c)$

R10 Commutativity of $\odot$;

$$\begin{aligned} a \odot b &= ab + a + b \\ &= ba + b + a \\ &= b \odot a \end{aligned}$$

So $a \odot b = b \odot a$

Therefore, R is a commutative ring with identity.

$\square$

b)
By Definition 2.8 HB Chapter 12 p54, we prove R is an integral domain;

Proof.

$$From part a), we know R is a commutative ring with identity.$$

No zero divisors;

Let $a, b \in R$ such that $a \odot b = 0$, then

$$ab + a + b = 0$$

adding 1 to both sides,

$$ab + a + b + 1 = 1$$

Factorising the LHS,

$$(a + 1)(b + 1) = 1$$

Since $a, b \in \mathbb{Z}$, the only way for the product to be 1 is if both $(a + 1) = 1$ and $(b + 1) = 1$

$$a + 1 = 1$$

$$b + 1 = 1$$

Therefore,

$$a = 0$$

$$b = 0$$

So the only way for $a \odot b = 0$ is if $a = 0$ or $b = 0$

Therefore, R has no zero divisors, and is an integral domain. $\square$

c)
By Definition 1.14 HB Chapter 11 pg48, we have already shown **R1-R10**, so we also need to find a multiplicative inverse for every non-zero element in R.

Multiplicative inverse;

Let $a \in R$ and $a \neq 0$, and let h be the multiplicative inverse of a, then

$$a \odot h = 0$$

$$ah + a + h = 0$$

$$ah + h = -a$$

$$h(a + 1) = -a$$

$$h = \frac{-a}{a + 1}$$

So the multiplicative inverse of a is $\frac{-a}{a+1}$

However, for h to be in R, $\frac{-a}{a+1}$ must be an integer, which is not the case for all $a \in R$

For example, if $a = 1$, then $h = \frac{-1}{2} \notin \mathbb{Z}$

Therefore, not every non-zero element in R has a multiplicative inverse, and so R is not a field.